PROJECT: SMART LIBRARY REQUEST WORKFLOW USING SERVICENOW

TEAM MEMBERS :

Jeevidha S

Sangeetha C

Kalpana S

Kanimozhi N

1. INTRODUCTION :

1.1 Project Overview :

The Smart Library Request Workflow is a system developed to automate the process of managing library service requests using the ServiceNow platform. In many educational institutions, library requests such as borrowing books or requesting new books are handled manually through paper forms or direct communication with library staff.

This manual process often causes delays and makes it difficult to track the status of requests. To solve this problem, the Smart Library Request Workflow system provides an online platform where users can easily submit library requests.

The system processes the request automatically using ServiceNow workflows and stores the request details in database tables. Library administrators can review the requests and approve or reject them through the system.

This automated system helps reduce manual work, improves request tracking, and provides faster and more efficient library services.

1.2 Purpose :

The main purpose of this project is to automate the library request process using the ServiceNow platform. In many educational institutions, library requests are still handled manually through paper forms or direct communication with library staff. This manual method can lead to delays and difficulties in managing multiple requests.

This project aims to provide an online system where students and staff can easily submit their library requests through a digital form. By using the ServiceNow platform, the system can automatically process and manage these requests in an organized manner.

Another purpose of this project is to reduce manual paperwork and improve the efficiency of library management. The automated workflow helps the library administrator to review and approve or reject requests quickly.

Overall, the system helps improve request tracking, reduce delays, and provide faster and more efficient library services for both users and administrators.

---

2. IDEATION PHASE :

2.1 Problem Statement:

In many educational institutions, library requests such as borrowing books or requesting new books are managed manually through paper forms or direct communication with library staff. This manual process is time-consuming and difficult to track.

Students may face delays in getting approvals, and library staff may find it difficult to manage multiple requests efficiently.

Therefore, there is a need for an automated system that allows users to submit library requests online and enables library administrators to manage requests efficiently.

The Smart Library Request Workflow system solves this problem by automating the request process using the ServiceNow platform.

2.2 Empathy Map Canvas :

Users:

Primary users of the system are Students and Library Staff who interact with the library services.

Users Think:

Users think that the current manual process for requesting books takes more time and effort. They want a simple and quick system to submit and track their library requests.

Users Feel:

Users often feel frustrated when there are delays in the manual request process. They expect a faster and more convenient way to request books and receive updates.

Users Say:

Users say that they need an easy method to request books without visiting the library repeatedly. They also want to know the status of their request.

Users Do:

Currently, users submit book requests through paper forms or directly communicate with the library staff. Sometimes they need to visit the library multiple times to check the request status.

Expected Solution:

The Smart Library Request Workflow system provides an online platform where users can submit requests easily and receive notifications about the request status.

2.3 Brainstorming :

During the brainstorming phase, the project team discussed several ideas to improve the library request process.

• Creating an online form for students to submit book requests easily.

• Automating the request approval process using workflows.

• Storing request details in a database for easy tracking.

• Sending notifications to users about request status updates.

• Reducing manual paperwork in library management.

• Improving communication between users and library administrators.

• Generating reports to monitor library request activities.

After evaluating these ideas, the team decided to develop a **Smart Library Request Workflow system using the ServiceNow platform** to automate and manage library requests efficiently.

---

3. REQUIREMENT ANALYSIS :

3.1 Customer Journey Map :

The Customer Journey Map describes the steps followed by the user while interacting with the Smart Library Request Workflow system.

1. User logs into the ServiceNow platform.
2. User opens the Smart Library Request application.
3. User fills the library request form with required details.
4. User submits the book request through the form.
5. The system stores the request details in ServiceNow tables.
6. The workflow automatically sends the request to the library administrator.
7. The administrator reviews the request.
8. The administrator approves or rejects the request.
9. The system updates the request status.
10. The user receives a notification about the request status.

3.2 Solution Requirements :

Functional Requirements :

• The system should allow users to submit library requests.
• The system should store request details in a database table.
• The system should automate the approval workflow.
• The system should notify users about request status.
• The system should allow administrators to manage requests.

Non-Functional Requirements :

• The system should be user-friendly.
• The system should securely store request data.
• The system should provide reliable performance.

3.3 Data Flow Diagram :

Input:
User submits library request details through the form.

Process:
ServiceNow workflow processes the request and sends it to the administrator.

Storage:
Request information is stored in ServiceNow database tables.

Output:
Request approval/rejection and notification sent to the user.

3.4 Technology Stack :

The Smart Library Request Workflow system is developed using the following technologies:

• **ServiceNow Platform**

Used as the main cloud platform to develop and manage the application.

• **ServiceNow Tables**

Used to store library request details such as user information, request type, and request status.

• **Forms**

Used to collect library request information from users through an online interface.

• **Flow Designer**

Used to automate the workflow process such as sending requests to administrators for approval.

• **Notifications**

Used to send updates to users about the status of their library requests.

---

4. PROJECT DESIGN :

4.1 Problem Solution Fit :

In many educational institutions, library requests are handled manually through paper forms or direct communication with library staff. This manual process often causes delays, difficulty in tracking requests, and increased workload for library administrators.

The Smart Library Request Workflow system provides an effective solution to this problem by automating the request process using the ServiceNow platform. Users can submit their library requests through an online form, and the system automatically processes the request using workflows.

The request is then sent to the library administrator for review and approval. The administrator can easily approve or reject the request, and the user receives a notification about the request status.

By implementing this automated system, the library can manage requests more efficiently, reduce manual effort, and improve the overall user experience.

4.2 Proposed Solution :

The proposed solution is to develop a Smart Library Request Workflow system using the ServiceNow platform to automate the library request process. Instead of using manual methods such as paper forms, users can submit their library requests through an online form available in the system.

Once the request is submitted, the system stores the request details in ServiceNow tables and processes the request using automated workflows. The workflow sends the request to the library administrator for review and approval.

The administrator can easily approve or reject the request through the system interface. After the decision is made, the user receives a notification about the request status.

This automated solution helps in reducing manual work, improving request tracking, and providing faster and more efficient library services.

4.3 Solution Architecture :

The Solution Architecture describes how different components of the Smart Library Request Workflow system work together to process library requests.

Components:

• **User Interface (Forms)**

Students and staff submit library requests through an online form available in the ServiceNow application.

• **Application Layer (ServiceNow Platform)**

The ServiceNow platform manages the application logic and processes user requests.

• **Workflow Automation (Flow Designer)**

Flow Designer automatically routes the request to the library administrator for approval.

• **Database (ServiceNow Tables)**

All request details such as user information, request type, and request status are stored in ServiceNow tables.

• **Admin Panel**

Library administrators review and approve or reject the submitted requests.

• **Notification System**

The system sends notifications to users informing them about the request status.

5. ADVANTAGES & DISADVANTAGES :

Advantages :

• Automates the library request process.

• Reduces manual paperwork and human errors.

• Provides faster request processing and approval.

• Makes it easy to track the status of library requests.

• Improves efficiency of library management.

• Saves time for both students and library staff.

Disadvantages:

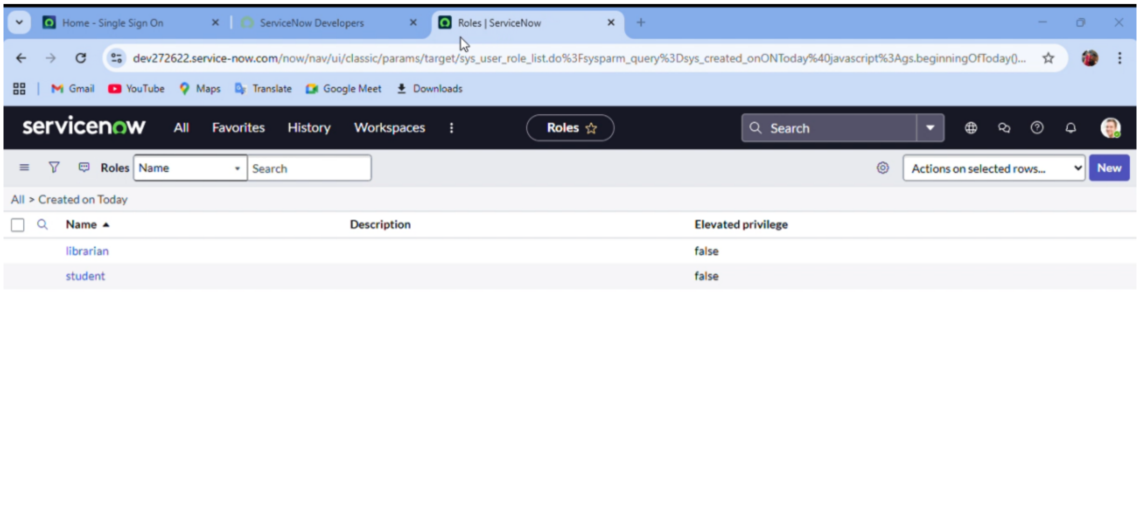
• Requires internet connection to access the system.

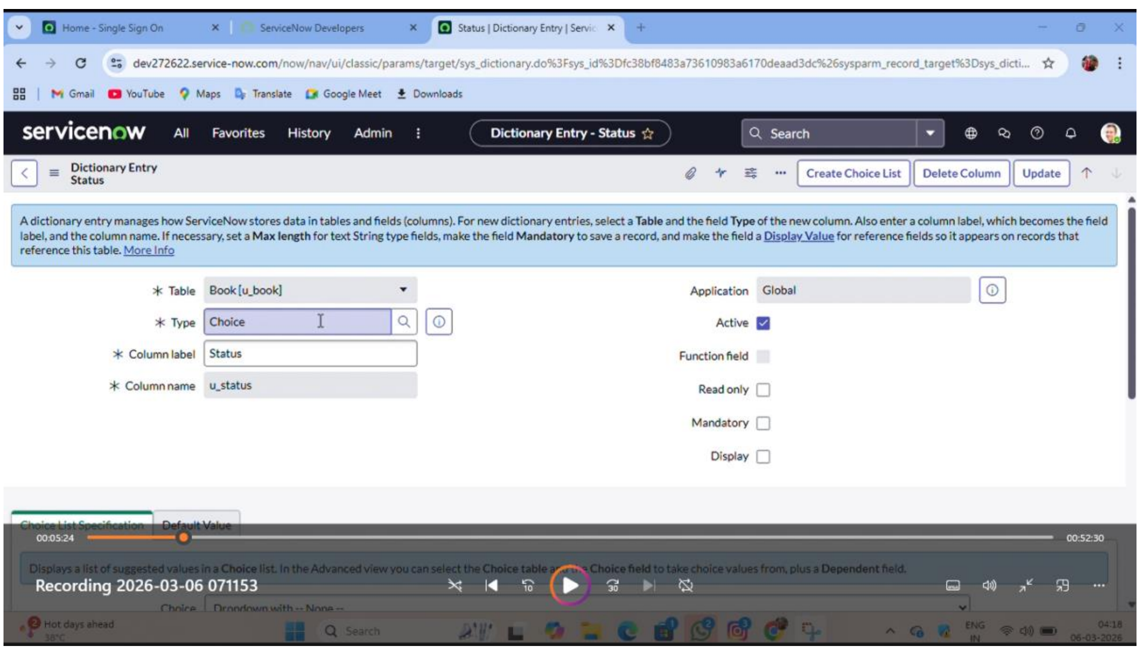
• Initial setup may require technical knowledge.

• Depends on the availability of the ServiceNow platform.

• Users need basic knowledge to use the online system.

6.SCREENCHOT :





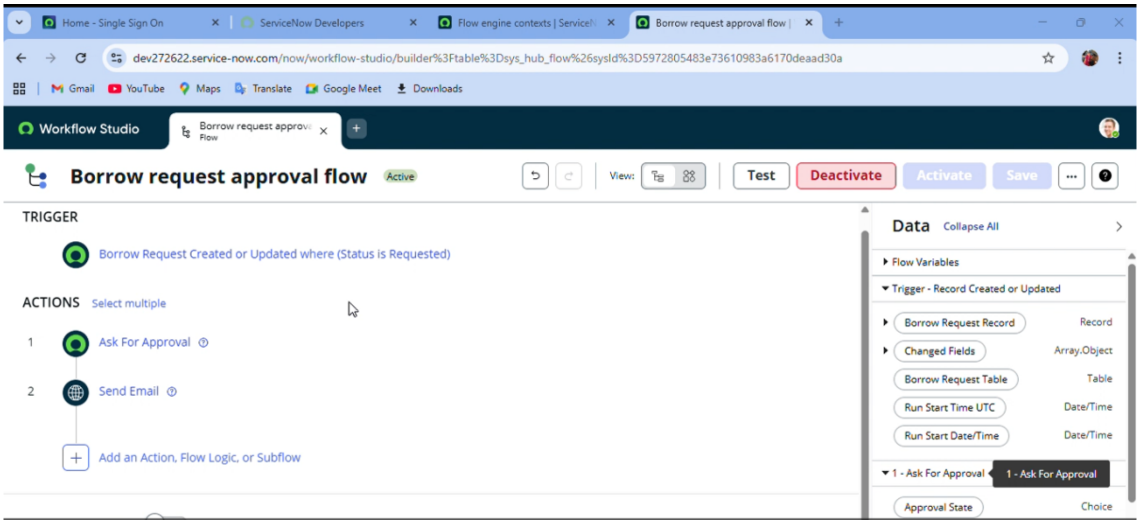
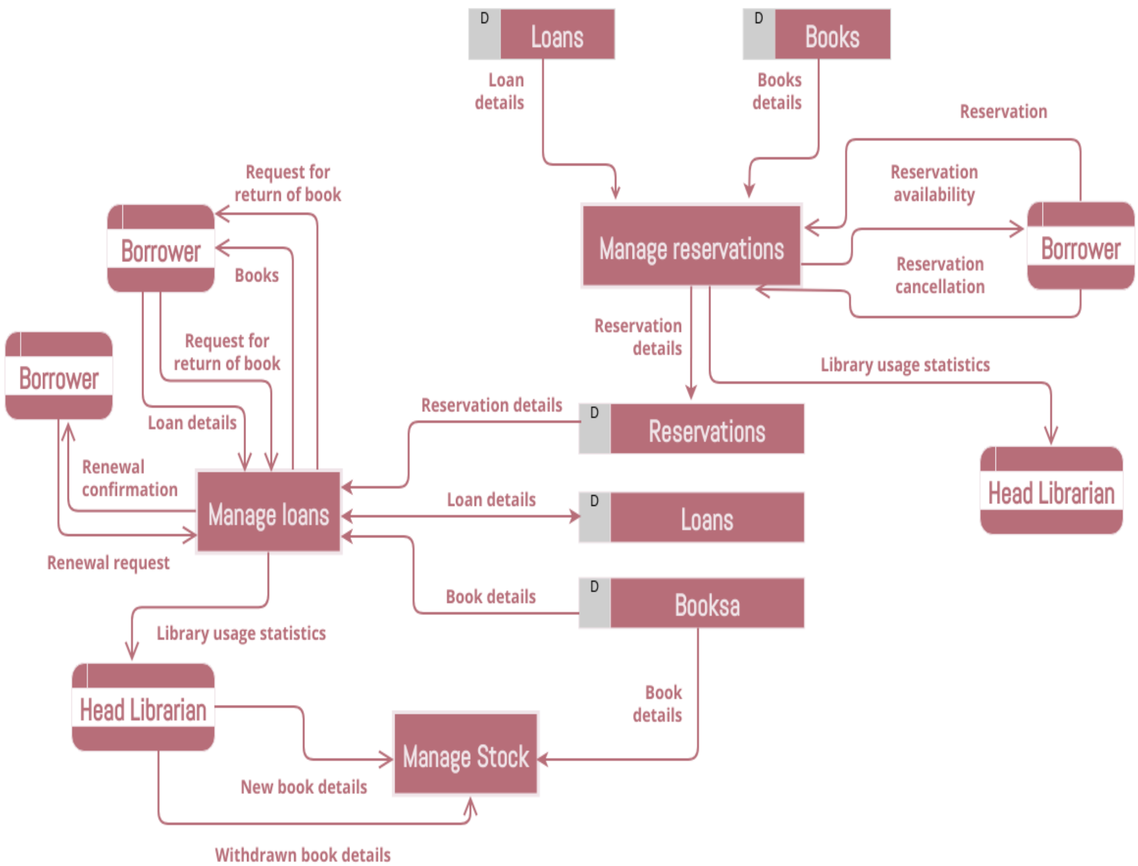

Home - Single Sign On
ServiceNow Developers
Flow engine contexts | ServiceN
Borrow request approval flow |
dev272622.service-now.com/now/workflow-studio/builder%3Ftable%3Dsys_hub_flow%26sysId%3D5972805483e73610983a6170deaad30a
Gmail
YouTube
Maps
Translate
Google Meet
Downloads
Workflow Studio
Borrow request approva
Flow
Borrow request approval flow
Active
View:
Test
Deactivate
Activate
Save
TRIGGER
Borrow Request Created or Updated where (Status is Requested)
ACTIONS
Select multiple
1
Ask For Approval
2
Send Email
Add an Action, Flow Logic, or Subflow
Data
Collapse All
Flow Variables
Trigger - Record Created or Updated
Borrow Request Record
Record
Changed Fields
Array.Object
Borrow Request Table
Table
Run Start Time UTC
Date/Time
Run Start Date/Time
Date/Time
1 - Ask For Approval
1 - Ask For Approval
Approval State
Choice

DIAGRAM:



D
Loans
D
Books
Loan details
Books details
Reservation
Request for return of book
Reservation availability
Borrower
Manage reservations
Borrower
Books
Reservation cancellation
Request for return of book
Reservation details
Borrower
Library usage statistics
Reservation details
Loan details
D
Reservations
Head Librarian
Renewal confirmation
Manage loans
Loan details
D
Loans
Renewal request
Book details
D
Booksa
Library usage statistics
Book details
Head Librarian
Manage Stock
New book details
Withdrawn book details

7. CONCLUSION :

The **Smart Library Request Workflow** system helps in improving the efficiency of library request management by automating the entire request process using the ServiceNow platform. The system allows users to submit library requests through an online form, which reduces manual paperwork and simplifies the request process.

Through workflow automation, the requests are processed quickly and sent to the library administrator for approval. This helps in saving time and makes it easier to track and manage library requests.

Overall, the system improves the quality of library services by providing a faster, more organized, and user-friendly method for handling library requests.

---

8. FUTURE SCOPE :

The Smart Library Request Workflow system can be further improved by adding more advanced features in the future.

• Book availability checking can be added so users can see whether a book is available in the library.

• The system can be integrated with digital library systems for better resource management.

• Automated book issue and return management can be implemented.

• Reporting and analytics features can be added to analyze library usage.

• A mobile application can be developed so users can submit requests through their smartphones.